(iii) $\int x^3 e^{-x}\,dx.$

Applying Bernoulli's formula

$$\int u\,dv = uv - u'v_1 + u''v_2 - \cdots$$

$$\int x^3 e^{-x}\,dx = (x^3)(-e^{-x}) - (3x^2)(e^{-x}) + (6x)(-e^{-x}) - (6)(e^{-x}) + c$$

$$= -x^3 e^{-x} - 3x^2 e^{-x} - 6xe^{-x} - 6e^{-x} + c.$$

	$dv = e^{-x}dx$
$u = x^3,$	$v = -e^{-x}$
$u' = 3x^2,$	$v_1 = +e^{-x}$
$u'' = 6x,$	$v_2 = -e^{-x}$
$u''' = 6,$	$v_3 = e^{-x}$

EXERCISE 11.7

Integrate the following with respect to x:

(1) (i) $9xe^{3x}$ (ii) $x\sin 3x$ (iii) $25xe^{-5x}$ (iv) $x\sec x\tan x$

(2) (i) $x\log x$ (ii) $27\,x^2e^{3x}$ (iii) $x^2\cos x$ (iv) $x^3\sin x$

(3) (i) $\dfrac{x\sin^{-1}x}{\sqrt{1-x^2}}$ (ii) $x^5e^{x^2}$ (iii) $\tan^{-1}\left(\dfrac{8x}{1-16x^2}\right)$ (iv) $\sin^{-1}\left(\dfrac{2x}{1+x^2}\right)$

11.7.8 Integrals of the form (i) $\int e^{ax}\sin bx\,dx$ (ii) $\int e^{ax}\cos bx\,dx$

The following examples illustrate that there are some integrals whose integration continues forever. Whenever we integrate function of the form $e^{ax}\cos bx$ or $e^{ax}\sin bx$, we have to apply the Integration by Parts rule twice to get the similar integral on both sides to solve.

Result 11.1

(i) $$\int e^{ax}\sin bx\,dx = \frac{e^{ax}}{a^2+b^2}[a\sin bx - b\cos bx] + c$$

(ii) $$\int e^{ax}\cos bx\,dx = \frac{e^{ax}}{a^2+b^2}[a\cos bx + b\sin bx] + c$$

Proof : (i)

$$\text{Let } I = \int e^{ax}\sin bx\,dx$$

$$\text{Take } u = \sin bx; du = b\cos bx\,dx$$

$$dv = e^{ax}; v = \frac{e^{ax}}{a},$$

Applying Integration by parts, we get

$$I = \frac{e^{ax}}{a}\sin bx - \int \frac{e^{ax}}{a} b\cos bx\,dx$$

$$I = \frac{e^{ax}}{a}\sin bx - \frac{b}{a}\int e^{ax}\cos bx\,dx$$

$$\text{Take } u = \cos bx; du = -b\sin bx\,dx,$$

$$dv = e^{ax}; v = \frac{e^{ax}}{a},$$

Again applying integration by parts, we get

$$I = \frac{e^{ax}}{a}\sin bx - \frac{b}{a}\left[\frac{e^{ax}}{a}\cos bx + \int \frac{e^{ax}}{a} b\sin bx\,dx\right]$$

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